



C23-CM-WD-405

23516

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DCME – FOURTH SEMESTER EXAMINATION

COMPUTER NETWORKS AND CYBER SECURITY

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. List any three advantages of optical fiber.
2. Write any three differences between LAN and WAN networks.
3. What is the purpose of ARP and RARP protocols?
4. List the classes in classful addressing and define the application of each class.
5. Define the terms passive and active attacks.
6. List the various security services in network security.
7. List five ingredients of a symmetric cipher.
8. Define steganography.
9. List any three security threats.
10. Define the terms virus and worm.

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[Contd...

PART—B

10×5=50

- Instructions :** (1) Answer *any five* questions.
(2) Each question carries **ten** marks.
(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. Define topology. Explain four basic types of network topologies,
12. Explain working of Simple Network Management Protocol (SNMP).
13. List and explain security goals of a network security.
14. Explain encryption and decryption process using rail fence transposition cipher with an example.
15. List and briefly define types of cryptanalytic attacks.
16. Encrypt the message '**AP SBTET MANGALAGIRI**' using a Caesar cipher. Ignore the spaces between words. Assume key 'Encryption'.
17. Define intrusion detection system. Explain how it works.
18. Explain the effectiveness of antivirus software in protecting systems from various types of malwares.

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